Isolation Level in Transaction :-

It tells, how the changes made by one transaction are visible to other transactions running parallel.

```
Transactional(propagation = Propagation.REQUIRED, isolation = Isolation.
ublic void updateUser(){                                  READ_COMMITTED)
// Some Operation here
}
```

Default isolation level, depends upon the DB which we are using

Like Most relational Database uses READ_COMMITTED as de default isolation, but again depends upon DB to DB

Dirty Read Problem

Transaction A reads the un-Committed data of other transaction. and
if other transaction get ROLLED BACK, the un-committed data which is read by Transaction A is known as Dirty Read

Concept && Coding

Time	Transaction A	Transaction B	DB status.
T_1	BEGIN-TRANSACTION	BEGIN TRANSACTION	id : 123 Status : free
T_2		Updat Row id : 123 Status = booked	id : 123 Status : booked (Not Committed by Transaction B yet)
T_3	Read Row id : 123 (Got Status = booked)		id : 123 Status : booked (Not Commited by Transaction B yet)
T_4		Rollback	id : 123 Status : Free (Un-committed Changes of Tx B got Rolled Back

Concept && Coding

Non-Repeatable Read Problem

If suppose Transaction A, reads the same row servel times and there is a chance that it get different value, Then its known as Non-Repeatable Read problem.

	Transaction A	DB
T_1	BEGIN-TRASACTION	ID : 1 Status : Free
T_2	Read Row ID : 1 (reads status : Free)	ID : 1 Status : Free
T_3		ID : 1 Status : Booked ← Some other Transaction changed and Committed the changes.
T_4	Read Row ID : 1 (reads status : Booked)	ID : 1 Status : Booked
T_5	COMMIT	

Phantom Read Problem

If suppose Transaction A executes sam query servel Times but there is a chance that rows returned are different. Than its known as Phantom Read Problem.

Concept && Coding

DB locking Types

Locking make sure that, no other transaction update the locked row.

Lock Type	Another shared lock	Another Exclusive l
Have shared lock	YES	No
Have Exclusive lock	No	No

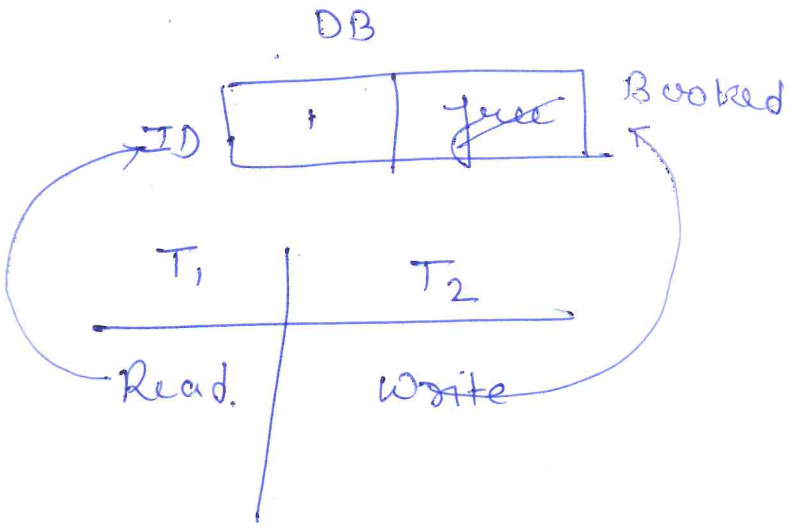
- Shared lock (S) also knows as READ lock.
- Exclusive lock (X) also knows as WRITE lock.

Concept && Coding

Isolation level and it's Type

Isolation level	Locking Strategy
Read Uncommitted	Read : No Lock acquired Write : No Lock acquired.

* Concurrency will be very high.



Isolation Level	Locking Strategy
Read Commited	Read : Shared Lock acquired and Released as soon as Read is done Write : Exclusive Lock acquired and keep till the end of transaction.

(f)
(S) ID: 1 free
T_1

	T_1	T_2
T_1	(S) ID: 1 Read: free	
T_2	(X) ID: 1 Status: Booked	
T_3		Read

Concept && Coding

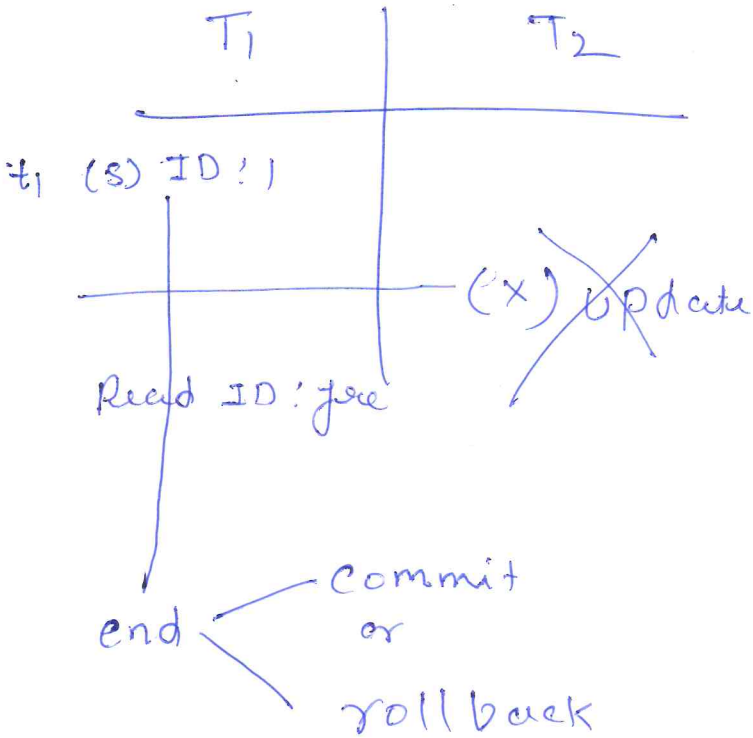
Isolation Level

Repeatable Read

Locking Strategy

Read :- Shared Lock acquired and Released only at the end of Transaction.

Write :- Exclusive Lock acquired and Released only at the end of Transaction.



DB

1	free

Concept && Coding

ISOLATION-LEVEL

Serializable

Locking strategy

Same as Repetable Read Locking strategy + apply Range Lock and lock is release only at the end of the Transaction.

	T_1	T_2
Time:	ID > 0 & ID < 5	
		Begin Txn
		(x) Insert ID = 2
	end tr	

	T_1	
(S)	1	free
(S).	4	

Range Lock [1, 2, 3, 4]

Concept && Coding